

I. PH4207 PROBLEM SET: II (TOTAL MARKS: 2FRM[O]-0)

Due on 18th March 2026

Q1. Consider the pure state

$$|\psi^{AB}\rangle = \frac{1+\sqrt{6}}{2\sqrt{6}} |00\rangle + \frac{1-\sqrt{6}}{2\sqrt{6}} |01\rangle + \frac{\sqrt{2}-\sqrt{3}}{2\sqrt{6}} |10\rangle + \frac{\sqrt{2}+\sqrt{3}}{2\sqrt{6}} |11\rangle$$

where the first qubit belongs to Alice and second to Bob.

1. Calculate $\rho^{AB} \rightarrow$ verify the purity of this state. [2]
2. Compute $\rho_A = \text{Tr}_B(|\psi\rangle\langle\psi|)$ & $\rho_B = \text{Tr}_A(|\psi\rangle\langle\psi|)$ [2]
3. Find the Schmidt decomposition of $|\psi\rangle$ [2]

Q2. For the following state, perform the singular value decomposition and examine the entanglement of the state,

$$|\psi^{AB}\rangle = \frac{1}{2} [|0\rangle_A |0\rangle_B - |0\rangle_A |1\rangle_B + |1\rangle_A |0\rangle_B + |1\rangle_A |1\rangle_B] \quad [4]$$